MIC-501P PRACTICAL

MICROBIOLOGY

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CURRENT OBJECTIVES

- Death rate is equal to growth rate. ***Stationary phase***
- Temperature of autoclave: ***121***
- For open manipulation of bacteria ***laminar air flow cabinet*** is used
- For killing bacterial endospore which method is best: ***Steam (autoclaving)***
- Which is used as mordant in Gram Staining....***Iodine***
- Result of catalyse test.... ***bright pink***
- CFU is the unit of.... ***viable bacterial***
- Which biosafety level are least dangerous to human health....***BSL 4***
- Why mirror is used in light microscope.... ***transmits light from a light source***
- What is the indication of antibiotic effect...***zone of inhibition***
- The primary role of illuminator....***lightning the slide***
- How many safety levels...***4***
- Lauria broth was first made by...***Bertani***
- Margins of bacteria except....***regular***
- Least reactive biosafety level***..... Level 1***
- Moist heat method ..... ***Autoclave***
- Reagent of indole test..... ***Kovacs***
- CFU unit for..... ***Viable bacteria***
- Particular type of bacteria separates in which medium____***Solid medium***
- Device used for separating different fraction..... ***centrifuge***
- BSL having types***...4***
- Generation time of bacteria . ***several hours***
- Temperature incubation in bacterial media..... ***37^0C***
- Regent of indole test ..... ***Kovac's or Ehrlich's***
- CFU is a measurement of____***viable cells.***
- Which instruments is used to separate fractions--- ***centrifuge.***
- Number of BSL levels---***4.***
- What is the purpose of mirror---- ***transmit slide from light source.***
- Molds are identified on the basis of ---***morphology.***
- LB is a standard medium used to grow bacteria--***E.coli.***
- Turbidity is determined by using an instrument--- ***spectrophotometer.***
- Urea test results color___ phenol red from its original orange yellow color to ***bright pink***

CURRENT SUBJECTIVES

Write any three types of light microscope

1. **Bright-field Microscope:** Illuminates specimens against a bright background.
2. **Phase-contrast Microscope:** Enhances contrast for transparent specimens.
3. **Fluorescence Microscope:** Utilizes fluorescence to visualize specific structures in cells or tissues.

How autoclaving contribute in LB broth preparation.

Autoclaving in LB broth preparation ensures sterility by subjecting the liquid medium to high-pressure saturated steam. This eliminates contaminants, providing an aseptic environment for bacterial culture growth, crucial for experiments requiring uncontaminated bacterial cultures in laboratory settings.

Write a note on BSL 3.

Biosafety Level 3 (BSL-3) Overview: BSL-3 encompasses pathogens causing severe diseases, posing high individual risk. While not easily transmitted, they demand strict containment measures due to their potential economic, health, and safety impacts.

Procedures and Requirements:

1. Rigorous Staff Training
2. Visible Standard Operating Procedures
3. Essential Equipment: Autoclave, Certified HEPA Filtered Class II Biological Safety Cabinet
4. Controlled Environment: Activities restricted to biological safety cabinets or suitable containment devices.

Containment Focus: Emphasizes personal and environmental safety, highlighting the critical need for comprehensive training, equipment, and controlled laboratory practices.

Why sterilization methods are crucial in healthcare. And consequences of infective methods of sterilization

Importance of Sterilization in Healthcare:

Sterilization methods are vital in healthcare to ensure the elimination of all microorganisms and their spores from medical instruments, equipment, and environments. This is crucial for preventing the transmission of infections and safeguarding patient well-being. Proper sterilization is essential to maintain aseptic conditions in medical settings, reducing the risk of healthcare-associated infections (HAIs) and promoting overall patient safety.

Consequences of Ineffective Sterilization Methods:

1. **Infections Spread:** Inadequate sterilization can lead to the persistence of infectious agents, causing the spread of infections within healthcare facilities.
2. **Patient Complications:** Patients may experience complications, prolonged recovery, or additional health issues due to infections contracted from improperly sterilized equipment.
3. **Increased Healthcare Costs:** Treating infections resulting from ineffective sterilization can significantly increase healthcare costs for both individuals and healthcare institutions.

4. **Legal and Ethical Implications:** Infections caused by lapses in sterilization may result in legal consequences and ethical dilemmas, impacting the reputation of healthcare providers and institutions.
5. **Reduced Trust in Healthcare:** Failures in sterilization methods can erode trust in healthcare systems, affecting patient confidence in seeking medical care.

Proper sterilization practices are imperative for maintaining a safe and hygienic healthcare environment, preventing infections, and ensuring optimal patient outcomes.

Differentiate biosafety levels.

There are four primary Biosafety Levels, each with its own specific safety practices and personal protective equipment (PPE):

Biosafety Level 1 (BSL-1):

Agents: Work with agents that do not typically cause disease in healthy individuals.

PPE: Basic laboratory attire, such as lab coats, gloves, and safety glasses.

Examples: Non-pathogenic strains of E. coli, Bacillus subtilis.

Biosafety Level 2 (BSL-2):

Agents: Work with moderately hazardous agents that can cause human disease but are usually treatable.

PPE: Lab coats, gloves, eye protection, face shields, and respiratory protection if necessary.

Examples: Influenza virus, Salmonella, Hepatitis B virus.

Biosafety Level 3 (BSL-3):

Agents: Work with pathogens that can cause serious or potentially lethal diseases through respiratory transmission.

PPE: Full-body, air-supplied positive-pressure suits, double-gloves, respirators, and other specialized equipment.

Examples: Tuberculosis (TB), SARS-CoV-2 (COVID-19), Yellow fever virus.

Biosafety Level 4 (BSL-4):

Agents: Work with dangerous and exotic agents that pose a high risk of causing severe or fatal disease for which no treatments or vaccines exist.

PPE: Completely enclosed, positive-pressure, and air-supplied suits with rigorous decontamination procedures.

Examples: Ebola virus, Marburg virus.

It's important to note that the specific PPE requirements may vary depending on the laboratory's specific protocols and the agents being handled.

Cell morphology of bacteria:

Bacteria and Archaea are classified by direct examination with the light microscope according to their morphology and arrangement. The basic morphologies are spheres (coccus) and round-ended cylinders (bacillus). But there may be others such as helically twisted cylinders (spirochetes), cylinders curved in one plane (selenomonads) and unusual morphologies (such as the square, flat box-shaped cells of the archaean genus Haloquadratum). Arrangements include pairs, tetrads, clusters, chains and palisades.

Differential media:

Differential media, also known as indicator media, are specifically designed culture media that facilitate the distinction of different bacteria based on their colony color or other morphological characteristics. These media incorporate dyes or metabolic substrates, allowing the growth of

multiple microorganisms with visually distinguishable colonies, aiding in microbiological identification.

What is membrane filter:

Membrane filters, typically composed of cellulose derivatives like acetate or nitrate, possess fine structures and are employed in microbiology. An example is the MF-Millipore, a cellulose ester mixture. These filters, with a nominal pore size of 0.22 ± 0.02 µm, are utilized for sterilizing aqueous and oily solutions but not organic solvents.

Define agar plate, agar slant, agar stab, broth culture:

1. Agar Plate: Petri dish containing solidified culture medium.
2. Agar Slant: Test lube containing solidified culture medium at a slope for preparing cultures and preservation of pure cultures in the lab.
3. Agar Stab: Test tube containing solid, culture medium for growth of anaerobic organisms especially when sealed with oil or deep culture system.
4. Broth Culture : Growth of bacteria in liquid medium.

Define CFU:

A Colony-Forming Unit (CFU) is a quantifiable measure of viable bacterial cells capable of forming a colony under specific growth conditions. Often expressed as CFU/mL for liquids and CFU/g for solids, it reflects the number of actively growing/dividing cells in a sample.

How liquid sample can be filtered through syringe filter:

To filter liquid samples using syringe filters, a sterile syringe with the desired sample is employed. The disposable syringe filter, available in various pore sizes (commonly 0.22 microns), is carefully attached to the syringe. The sample is then pushed through the filter, and the filtrate is collected in a sterile falcon tube. This method allows for the removal of microbial contaminants, enabling the sterilization of thermolabile solutions while preventing the rupture of thicker Seitz filters.

Significant of media

Differentiate dark Field, bright filed, fluoricent filed.

- **Dark Field:** Illumination from the side, revealing light objects against a dark background.
- **Bright Field:** Standard illumination from below, providing a bright background.
- **Fluorescent Field:** Uses fluorescence to visualize specimens, highlighting specific structures with fluorochromes under ultraviolet light.

Grow and preserve L-Broth.

Luria broth (LB) is a nutrient-rich medium for bacterial culture. To grow, prepare LB with specific components and autoclave. Inoculate with a single colony, incubate at 37°C, and check for growth. For preservation, create a glycerol stock after bacterial culture incubation, allowing long-term storage. LB, devised by Bertani during phage lysis investigations, is crucial in microbiology for supporting bacterial growth and maintaining strains.

Significance of biosafety levels:

Biosafety levels are crucial for isolating dangerous biological agents in laboratories. They establish biocontainment precautions, protecting workers by implementing specific guidelines, engineering controls, and work practices. Adhering to these levels prevents accidental infections among researchers and ensures safety when handling organisms posing threats to human health during testing or research on clinical specimens.

Explain in detail the laminar air flow cabinet role in open culture manipulation.

The laminar air flow cabinet is a semi-closed cabinet designed for open culture manipulation in laboratories. It employs a filtered airflow system that creates a sterile work environment, preventing contamination during the handling of cultures. This specialized cabinet ensures a continuous flow of clean air, safeguarding both the laboratory personnel and the biological materials from external pollutants, ultimately maintaining aseptic conditions crucial for sensitive procedures in microbiology and cell culture work.

why inculating loop become flame before streaking process.

The inoculating loop is flamed before the streaking process to sterilize it by reaching a red-hot temperature. This eliminates any potential contaminants and ensures that only the desired microorganisms are transferred during the streaking procedure, maintaining the integrity of the culture being inoculated.

what is steak method

The steam method, commonly known as autoclaving, is a prevalent sterilization technique used for drugs and laboratory equipment. Operating under high pressure, steam at temperatures exceeding 100°C is employed in an autoclave. As pressure increases, water boils above 100°C, reaching 121°C at 15 pounds per square inch. This process effectively destroys bacterial spores, including their rigid protective covers, ensuring thorough sterilization. The autoclave functions like a pressure cooker, maintaining a temperature of 121°C for 15 minutes to achieve optimal sterilization results.

Difference between simple and differential stain

1. Simple stains techniques Simple staining is performed with basic dyes such as crystal violet or methylene blue.

2. Differential stains techniques Differential stain technique distinguishes two kinds of organisms. There are two types of this technique; one is Gram stain technique while other is acid fast technique.

How culture colony can be counted?

To count culture colonies, a plate count method, or spread plate technique, is employed. A ten-fold dilution of the bacterial culture is prepared, reaching a dilution factor of 10^-6. Aliquots of each dilution are spread on three agar plates using a sterile spreader. After incubation at 37°C for 24 hours, the plates with 30 to 200 colonies are chosen. The number of bacterial colonies on these plates is counted, and the estimated number of bacteria per milliliter is calculated using the formula B = N/d.

Dry heat method:

Dry heat methods involve subjecting substances to elevated temperatures without moisture. Techniques like flaming, incineration, hot air ovens, and radiation sterilization are used. Flaming and incineration expose instruments to flame, burning microbes. Hot air ovens sterilize dry materials at 160°C, killing spores and ensuring sterilization.

Different culture media:

1. Nutrient Agar: Supports general microbial growth.
2. Blood Agar: Contains blood, used for hemolysis studies.
3. MacConkey Agar: Selective for Gram-negative bacteria.
4. Sabouraud Agar: Favors fungal growth.
5. Mannitol Salt Agar: Selective for staphylococci.
6. Eosin Methylene Blue (EMB) Agar: Inhibits Gram-positives.
7. Thayer-Martin Agar: Selective for Neisseria species.
8. Chocolate Agar: Supports fastidious organisms.
9. Lowenstein-Jensen Medium: Used for Mycobacterium tuberculosis.
10. TCBS Agar: Selective for Vibrio species.

Why lag phase significant in growth of bacteria.

The lag phase is crucial in bacterial growth as it represents the period when bacteria adapt to the new environment, synthesize necessary components, and prepare for rapid division in the subsequent log phase. It ensures optimal conditions for exponential growth by laying the foundation for a robust and synchronized cellular machinery.

Write components of bacterial colony shape:

1. **Form**: Describing the shape of the colony, such as circular, irregular, filamentous, or rhizoid.
2. **Elevation**: Representing the side view of a colony, including flat, raised, umbonate, crateriform, convex, or pulvinate.
3. **Margin**: The edge of a colony, with examples like entire (smooth), irregular, undulate, lobate, curled, or filiform.
4. **Size**: Measured in millimeters or described as pin-point, small, medium, or large.
5. **Appearance**: Surface characteristics like shiny, smooth, dull, veined, rough, wrinkled, or glistening.

Ethylene oxide acts as a sterilizing agent write its advantages and disadvantages:

Advantages of Ethylene Oxide (EO) as a Sterilizing Agent:

1. Effective at low temperatures (20–60°C).
2. High penetration capacity, reaching through some plastic foils.
3. Suitable for sterilizing items sensitive to heat.

Disadvantages of Ethylene Oxide (EO) as a Sterilizing Agent:

1. Flammable, toxic, and a strong mucosal irritant.
2. Ineffective against dried microorganisms.
3. Requires a relative humidity level of 40–90% for optimal efficacy.
4. Slow desorption from plastics, rubber, and similar materials.

What is chemical sterilization?

Chemical sterilization involves using toxic gases like ethylene oxide or chlorine compounds to eliminate microorganisms on surfaces or equipment. These chemicals denature proteins, disrupt cell membranes, and coagulate essential substances, ensuring effective sterilization, though precautions for safety and cost are necessary.

what is gram staining?

Gram staining is a widely used laboratory technique in microbiology to differentiate and classify bacteria into two main groups based on their cell wall composition. The two groups are Gram-positive bacteria, which have a thick peptidoglycan layer in their cell walls, and Gram-negative bacteria, which have a thinner peptidoglycan layer along with an outer membrane containing lipopolysaccharides.

Procedure of gram staining stepwise?

Gram staining is a widely used laboratory technique in microbiology to differentiate and classify bacteria into two main groups based on their cell wall composition. The two groups are Gram-positive bacteria, which have a thick peptidoglycan layer in their cell walls, and Gram-negative bacteria, which have a thinner peptidoglycan layer along with an outer membrane containing lipopolysaccharides.

Here's a step-by-step guide to performing a Gram staining practical:

Materials Needed:

- Bacterial culture (pure culture of known bacteria)
- Microscope slides

- ❖ Bunsen burner or alcohol lamp
- ❖ Inoculating loop or sterile cotton swabs
- ❖ Crystal violet stain
- ❖ Gram's iodine solution (Gram's mordant)
- ❖ Ethanol or acetone (decolorizing agent)
- ❖ Safranin stain
- ❖ Microscope

Procedure:

1. Prepare a bacterial smear on a slide and heat-fix it.
2. Apply crystal violet stain, Crystal violet stains all the bacterial cells,
3. Rinse, then apply Gram's iodine solution. This helps to form a complex between the crystal violet and the bacterial cells, enhancing the retention of the stain.
4. Rinse again, then apply decolorizing agent (ethanol/acetone) briefly and rinse immediately.
5. Apply safranin stain which stains the decolorized cells, which in this case are the Gram-negative bacteria., rinse, and blot the slide dry.

Results:

Observe under a microscope: Purple = Gram-positive, Pink = Gram-negative.

Mode of action of chemical sterilization.

The modes of action in chemical sterilization involve protein coagulation, disruption of cell membranes leading to exposure or damage, removal of essential enzyme sulfhydryl groups, and substrate competition. These processes collectively ensure the effective sterilization of articles using toxic gases or chemicals

Indole test procedure, purpose and result.

1. **Inoculation:** Introduce the test organism into tryptophan broth.
2. **Incubation:** Allow the broth to incubate at 37°C for 24-28 hours in ambient air.
3. After incubation, add 0.5 ml of Kovac's reagent to the broth culture.

Purpose: The Indole Test assesses an organism's ability to break down tryptophan, differentiating Enterobacteriaceae and other genera. It is commonly employed in biochemical tests like IMVIC and SIM.

Result Interpretation:

- **Positive Result:** The addition of Kovac's reagent yields a pink-colored compound.
- **Negative Result:** No color change occurs post the addition of the reagent, indicating a negative result.